Unit: 6 Unit 7: Factoring
Topic: Sum and Difference of Cubes

Name: ________________
Date: ___________

Homework (Jalapeno)

Q1. The ancient Egyptians used $x^3 + y^3$ to calculate volumes. What is $4^3 + 5^3$?
(A) 189
(B) 207
(C) 225
(D) 243
(E) 261

Q2. The ancient Greeks recognized $a^3 - b^3 = (a-b)(a^2+ab+b^2)$. What is $8^3 - 2^3$ factored?
(A) $(8-2)(8^2+8*2+2^2)$
(B) $(8+2)(8^2-8*2+2^2)$
(C) $(8-2)(8^2-8*2+2^2)$
(D) $(8+2)(8^2+8*2+2^2)$
(E) $(8*2)(8^2+8*2+2^2)$

Q3. A Mayan artifact has a volume of $x^3 - 9^3$. What is the factored form?
(A) $(x-9)(x^2+9x+81)$
(B) $(x+9)(x^2-9x+81)$
(C) $(x-9)(x^2-9x+81)$
(D) $(x+9)(x^2+9x+81)$
(E) $(x-9)(x^2+9x-81)$

Q4. What is the value of $3^3 + 2^3$?
(A) 17
(B) 25
(C) 29
(D) 35
(E) 47

Q5. An ancient Chinese problem gives $16^3 - 9^3$. What is its factored form?
(A) $(16-9)(16^2+16*9+9^2)$
(B) $(16+9)(16^2-16*9+9^2)$
(C) $(16-9)(16^2-16*9+9^2)$
(D) $(16+9)(16^2+16*9+9^2)$
(E) $(16+9)(16^2-16*9-9^2)$

Q6. A historian finds $x^3 + 8^3$ in an ancient text. What is its factored form?
(A) $(x+8)(x^2-8x+64)$
(B) $(x-8)(x^2+8x+64)$
(C) $(x+8)(x^2+8x+64)$
(D) $(x-8)(x^2-8x+64)$
(E) $(x+8)(x^2-8x-64)$

Q7. The Rosetta Stone mentions $6^3 - 4^3$. What is its factored form?
(A) $(6-4)(6^2+6*4+4^2)$
(B) $(6+4)(6^2-6*4+4^2)$
(C) $(6-4)(6^2-6*4+4^2)$
(D) $(6+4)(6^2+6*4+4^2)$
(E) $(6-4)(6^2+6*4-4^2)$

Q8. An archaeologist discovers 9^3+x^3 in an ancient scroll. What is the factored form?
(A) $(9+x)(9^2-9x+x^2)$
(B) $(9-x)(9^2+9x+x^2)$
(C) $(9+x)(9^2+9x+x^2)$
(D) $(9-x)(9^2-9x+x^2)$
(E) $(9+x)(9^2-9x-x^2)$

Open Ended Questions (FRQ)

Q9. Factor $2x^3 + 16$ and show all steps.

Q10. A museum curator wants to display a cube with volume $x^3 - 64$. Factor the expression and find the value of x if the volume is 0. Show all steps and calculations.

Chef's Tips

- Tip 1: Emphasize recognizing and applying sum/difference of cubes formulas.
- Tip 2: Ensure students can factor expressions with variables and constants.
- Tip 3: Provide opportunities for students to practice factoring and simplifying expressions.